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Technical Business Analyst Assessment

Focus Areas: SQL & API Understanding

Assessment Overview

This assessment is designed to evaluate your practical and analytical skills as a **Technical Business Analyst**, with a focus on:

- Writing and optimizing SQL queries
- Interpreting business requirements into data insights
- Understanding API design and functionality
- Applying structured thinking to real-world technical scenarios

You will be assessed not only on correctness, but also on:

- Clarity of your approach
- Efficiency and scalability considerations
- Ability to communicate technical solutions

Submission Guidelines

- Share your solution via **MS Word**, **GitHub repository**
- Include a **README file** with:
  - Setup instructions
  - Technology stack used
  - Any assumptions made

Notes:

- *Clearly explain your reasoning for each solution*
- *Focus on both **technical accuracy** and **business relevance***
- *Structure your answers in a way that is easy for stakeholders to understand*
- *Where applicable, include diagrams, assumptions, and examples*

Assessment Structure

This assessment is divided into **two parts**:

- **Part 1:** SQL Case Study (Compulsory)
- **Part 2:** API Case Studies (Choose any 2 tasks)

Part 1: SQL Case Study

Introduction

This section evaluates your ability to:

- Translate business problems into SQL solutions
- Work with relational data structures
- Optimize queries for performance
- Derive meaningful insights from telecom data

You are expected to:

- Write SQL queries
- Clearly explain your logic and approach
- Highlight performance considerations (indexes, joins, etc.)

Telecom Related Case Study For SQL:

You are a senior SQL and data analysis professional in a telecom company. Below are three business cases you must address using SQL queries and analytical thinking. After each solution, you will explain your approach, ensuring efficiency, scalability, and correctness.

Table Structures:

1. **Customers**
   - Customer_ID (Primary Key)
   - Name
   - Signup_Date
   - Email

2. **Subscriptions**
   - Subscription_ID (Primary Key)
   - Customer_Id (Foreign Key)
   - Plan_Type (E.G., Basic, Premium)
   - Start_Date
   - End_Date

3. **Usage**
   - Usage_ID (Primary Key)
   - Customer_ID(Foreign Key)
   - Date
   - Data_Used_Mb
   - Call_Minutes

Use Case 1:

Sales Report Bottleneck
The company is experiencing slow performance when generating monthly revenue reports. Your task is to identify bottlenecks in the usage, subscription, and customer data. Write optimized SQL queries to extract monthly revenue by customer segment and plan type. Ensure you explain how your query uses indexed columns and efficient joins.

Use Case 2:

Churn Prediction
Identify customer segments at high risk of cancelling. Analyze usage (data and call minutes), subscription duration, and billing patterns. Write queries that flag at-risk customers based on declining usage or short subscription length.

Use Case 3:

Upsell Opportunities
Identify customers nearing their plan limits who may benefit from an upgrade. Analyze usage by plan type. Write queries to find these top candidates, grouping them by customer and highlighting those just under thresholds.

Part 2: API Case Studies (Select Any 2)

Introduction

This section evaluates your understanding of:

- API design and structure
- RESTful principles
- Data validation and error handling
- Integration between systems

You are not only expected to demonstrate technical knowledge, but also:

- How APIs support business processes
- How you would document and structure solutions as a Business Analyst

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Task 1: **Basic API Development with Swagger**	**Objective:** Develop a simple User Management API. **Requirements:** Implement the following endpoints: • POST /users – Create a user • GET /users/{id} – Retrieve user details • PUT /users/{id} – Update user • DELETE /users/{id} – Delete user • Integrate Swagger (OpenAPI) for API documentation • Use an in-memory data store or simple database
Task 2: **Validation and Error Handling**	**Objective:** Enhance the API with proper validations and error handling. **Requirements:** • Validate input fields (e.g., required fields, email format) • Implement appropriate HTTP status codes: • 400 – Bad Request • 404 – Not Found • 500 – Internal Server Error • Provide meaningful error responses in JSON format
Task 3: **Database Integration**	**Objective:** Persist data using a relational database. **Requirements:** • Integrate a database (e.g., MySQL, PostgreSQL, SQL Server) • Store and retrieve user data from the database • Provide database schema/script
Task 4: **Authentication (Optional but Recommended)**	**Objective:** Secure the API endpoints. **Requirements:** • Implement a login endpoint • Use JWT-based authentication • Protect user-related APIs
Task 5: **API Design (Swagger-First Approach)**	**Objective:** Design an API before implementation. **Requirements:** Create an OpenAPI (Swagger) specification for an Order Management system: • Create Order • Get Orders • Update Order Status

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